Poststreptococcal glomerulonephritis and IgA nephropathy:

Prof. Dr. Ruqaya M. Al Barzinji
Hawler Medical University
College of Medicine
Department: Microbiology
SGL

Glomerulonephritis

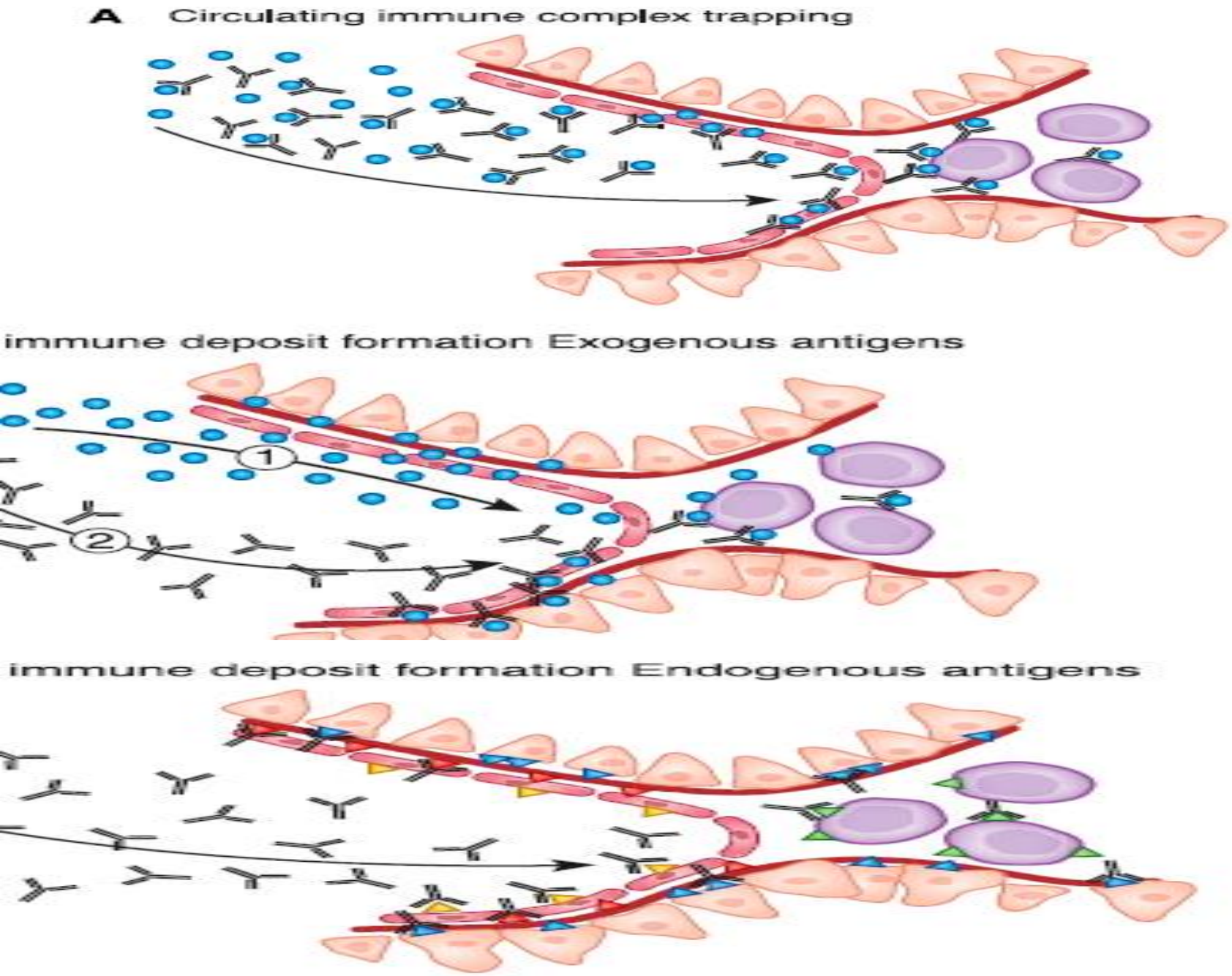
- Glomerulonephritis (GN (is a specific set of **renal diseases** in which an immunologic mechanism **triggers inflammation and proliferation** of glomerular tissue that can result in damage to the basement membrane, mesangium, or capillary endothelium of the kidney.

Immune mechanisms of glomerulonephritis

- **-1Deposition** of circulating **antigen-antibody complexes** within glomeruli (Iimmune complex nephritis(
- **-2Reaction** of **circulating antibodies with:**

A-Antigen that have been trapped their .

B-Antigen which are **part of the glomerular tissue**)**antiGBM-nephritis(.**


A Circulating immune complex trapping
immune deposit formation Exogenous antigens
1
2
immune deposit formation Endogenous antigens

Post-Streptococcal Glomerulonephritis(PSGN)

- **PSGN** is a kidney disease that can develop after a Group A streptococcal infection.
- PSGN is a result of the body's immune system fighting off the group A strep infection.
- PSGN usually occurs **10 days** after **strep throat** or **scarlet fever** and about **3 weeks** after a Group A streptococcal **skin infection**.
- Symptoms of PSGN include:
- **Dark, reddish-brown urine**
- **Swelling in the face, hands and feet**
- **Decreased amount of urin**e or decreased need to urinate
- **Fatigue**.
- Most people with PSGN recover within a few weeks after treatment, but in **rare** cases long-term kidney damage, as kidney failure, can occur.

Poststreptococcal glomerulonephritis

- Streptococcus pyogen is a Gram positive bacteria secretes an exotoxin called streptococcal pyogenic exotoxin B (**SpeB**(.
- **SpeB** has been implicated in most cases of epidemic **poststreptococcal glomerulonephritis** (PSGN)..
- **SpeB** is a cysteine protease ,which cleaves various host factors and function as a nephritogenic antigen.
- SpeB represents 90% of the secreted extracellular protein made by **nephritogenic strains** of group **A streptococci**.

- **Antibody** to SpeB correlates with **disease activity** in **PSGN.**
- **SpeB** exhibits **molecular mimicry** with **endothelial cells therefore s**ome **subendothelial deposits** are present in PSGN**.**
- **SpeB alone** can **activate complement** directly through the mannose binding lectin (**MBL)pathway** independent of IgG.
- SpeB also exhibits **plasmin-binding properties** that facilitate **complement activation** and might cause **proteolysis of glomerular basement membrane** (GBM).

2-IgA nephropathy

- **IgA nephropathy** is one of the common forms of **glomerulonephritis**, it is an autoimmune disease.
- **Immune-mediated damage** to the basement membrane caused by the deposition of IgA immunoglobulin in the glomerular basement membrane causing inflammation that damage kidney tissues and results:
- **Hematuria**
- **Proteinuria**
- **Renal deficiency**
- Berger was the first to describe the disease, so it also called
- **Berger disease**.

IgA nephropathy

- The diagnosis of IgA nephropathy is established by determination of IgA1 subclass as the dominant immunoglobulin in glomeruli that has **deficient in galactose on the hinge-region.**
- **The immunoglobulin is called Galactose deficient- IgA1(GD-IgA1).**

Etiology

- **Genetic and environmental** factors may play role in IgAN.
- **Gender:** IgA nephropathy incidence is typically diagnosed in young adults **males** more commonly than **females**.
- **Ethnicity:** The disease is more common in the whites than blacks.
- **Age:** The disease develops between the late teens and late thirteens.
- **Family history** also have a role in IgA nephropathy.
- **Infectious disease**(RT and GIT) infections preceding the nephropathy. The disease may also be seen in Celiac disease, Hepatitis, Cirrhosis, HIV.

Pathophysiology

- The **pathogenesis** of IgAN remains incompletely understood.
- There is a defect in galactosylation of serum IgA1 from patients with IgAN.
- This defect resulting in **reduced galactosylation** of the **O-linked glycans** of the IgA1 hinge region by reduced expression of intracellular enzyme galactosyltransferase that has been proposed as explanation for abnormal glycosylation of IgA.
- Circulatory Galactose deffecient-IgA1(GD IgA1) is recognized as an autoantigen and attacked by **IgG or IgA auto- abs**, resulting in the formation of **immune complexes.**

- Some of these circulating complexes containing **undergalactosylated IgA1** deposit in glomeruli and
- Activate mesangial cells by;
- **Modulate their functions**
- **Integrin expression**
- **Vascular endothelial growth factor synthesis.**
- **Induce glomerular injury** through cellular **proliferation and overproduction** of components of extracellular matrix and cytokines leading to the local overproduction **of cytokines, chemokines and complement activation, glomerular scleroses and interstitial fibroses.**

- Abnormal **physicochemical** properties of circulating IgA, as **size, charge, and glycosylation** may play a role in their deposition.
- The presence of antibodies bound to O-linked glycans in the IgA1 hinge region reduces the hepatic clearance of the circulating **immune complexes**, which results in **higher levels.**
- The disease is characterized by **mesangial deposits** of IgA, IgG, IgM and C3 .

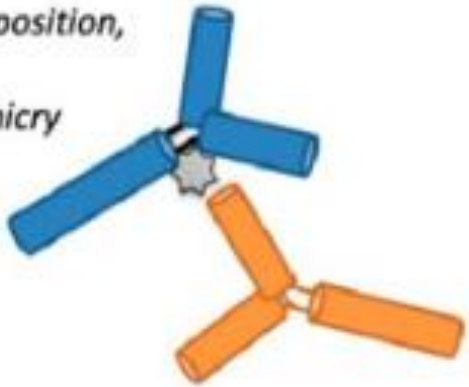
1. Increased circulating levels of Gd-IgA1

- *Genetic predisposition*
- *Mis-trafficking of B cells from mucosal to systemic sites*



2. Production of Anti-IgA1 antibodies (IgA or IgG)

- *Genetic predisposition, HLA haplotype*
- *Molecular mimicry*
- *Viral infection*
- *Food antigens*



3a. Immune complexes form in the circulation



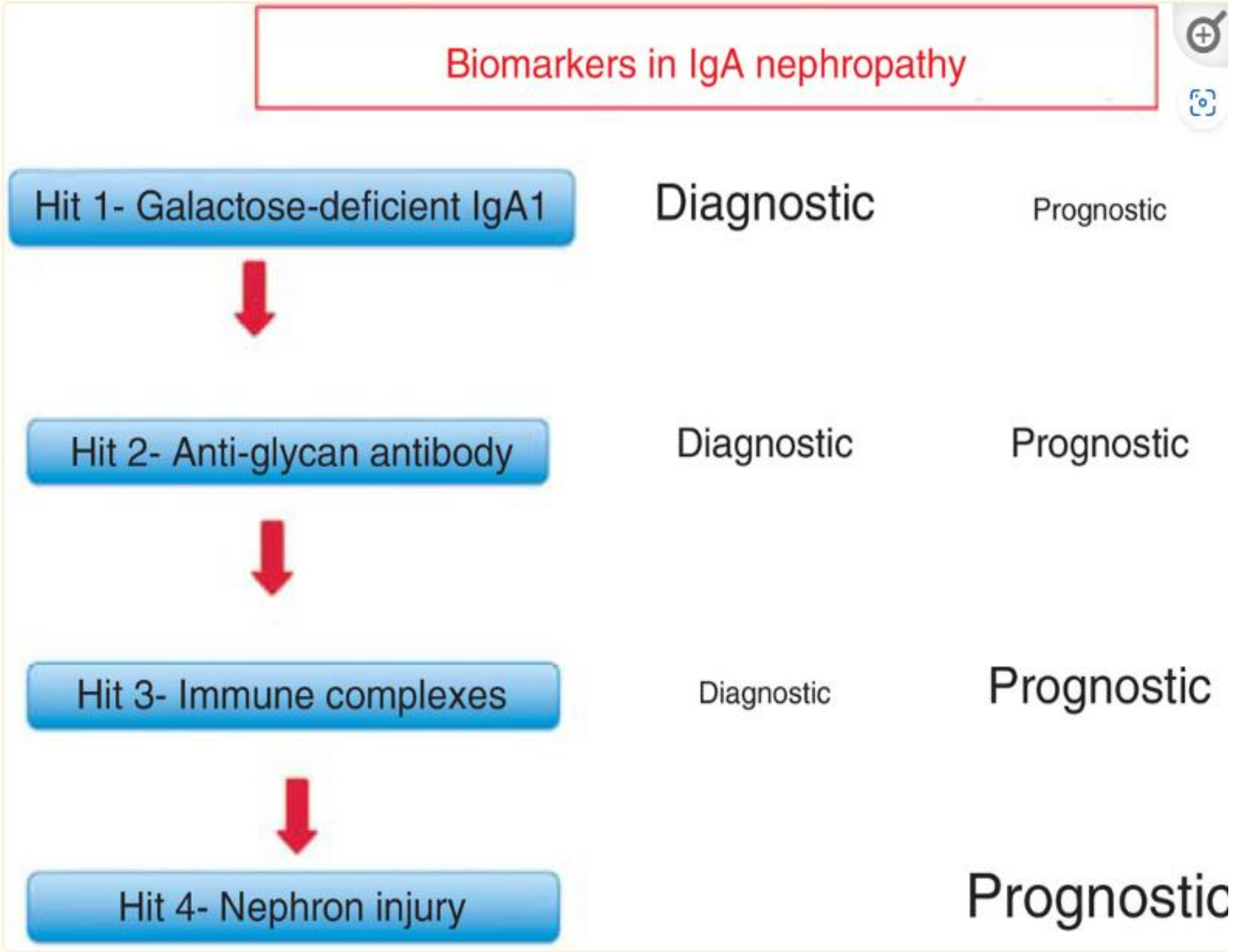
3b. Immune complexes form *in situ*



4. Immune complexes in the mesangium cause local immune activation & injury



- *Complement activation*
- *Cytokine/chemokine release*
- *Matrix production*
- *Mesangial proliferation*
- *Glomerular sclerosis*
- *Interstitial fibrosis*

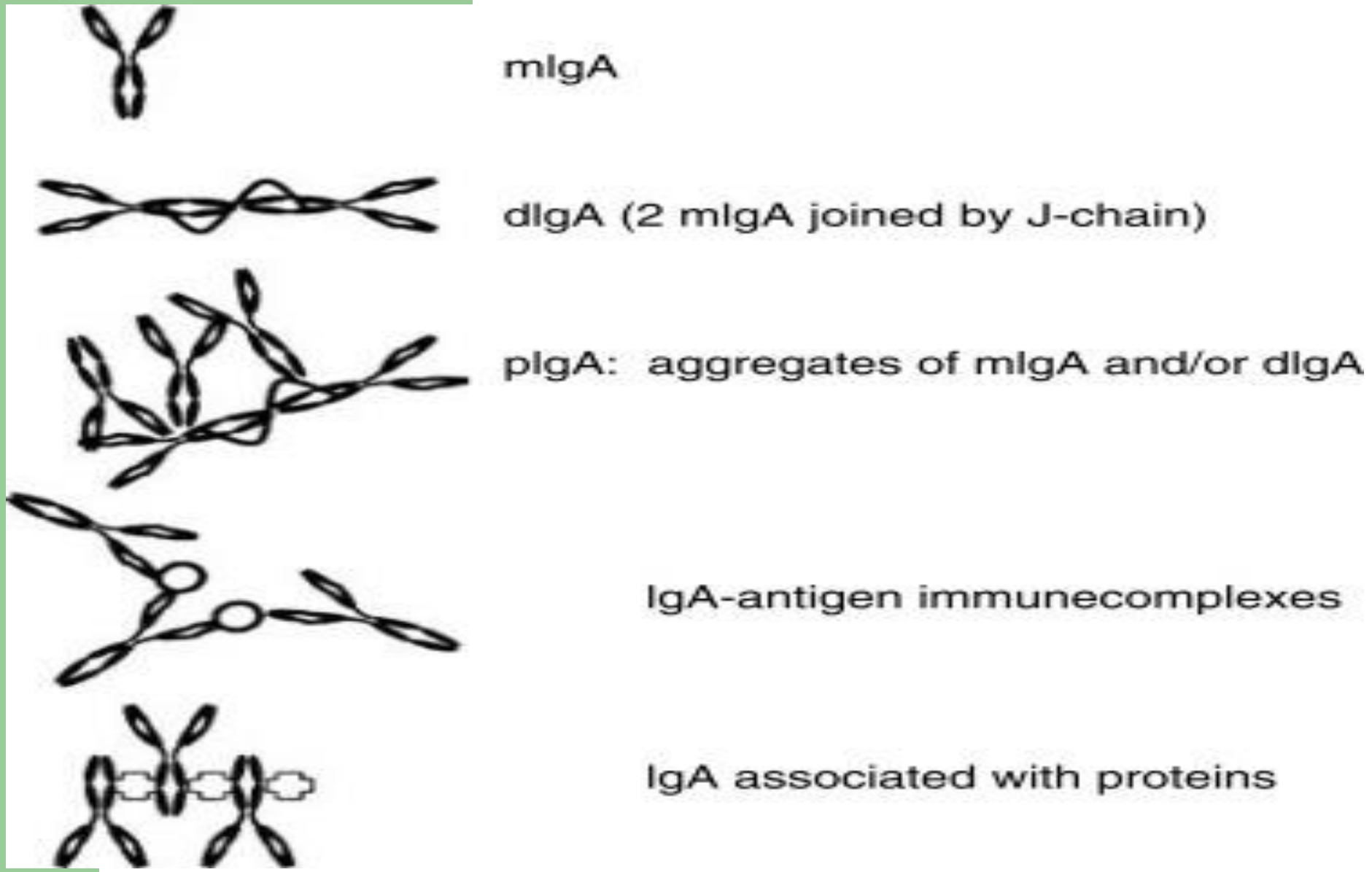

Biomarkers in IgA nephropathy
Hit 1- Galactose-deficient IgA1
Diagnostic
Prognostic
Hit 2- Anti-glycan antibody
Diagnostic
Prognostic
Hit 3- Immune complexes
Diagnostic
Prognostic
Hit 4- Nephron injury
Prognostic

IgA nephropathy symptoms

- Patients can present with a range of signs and symptoms, from **asymptomatic microscopic** haematuria to **macroscopic** haematuria.
- The clinical progression varies, with 30–40% of patients reaching end-stage renal disease 20–30 years after the first clinical presentation.
- About 40% of cases have recurrent episodes of macroscopic hematuria, frequently preceded 1 or 2 days earlier by **infections**.
- Mild **proteinuria** usually develops which may progress to heavy **proteinuria**.

IgA isotypic forms

- IgA occurs in two isotypic forms, IgA1 and IgA2.
- Circulating IgA is produced in the bone marrow and contains approximately 90% IgA1 and 10% IgA2.
- It is exists predominantly in the monomeric form with a small proportion as **macromolecular IgA**, which consists of dimeric IgA (dIgA) and polymeric IgA (pIgA).



Different forms of IgA in the circulation.

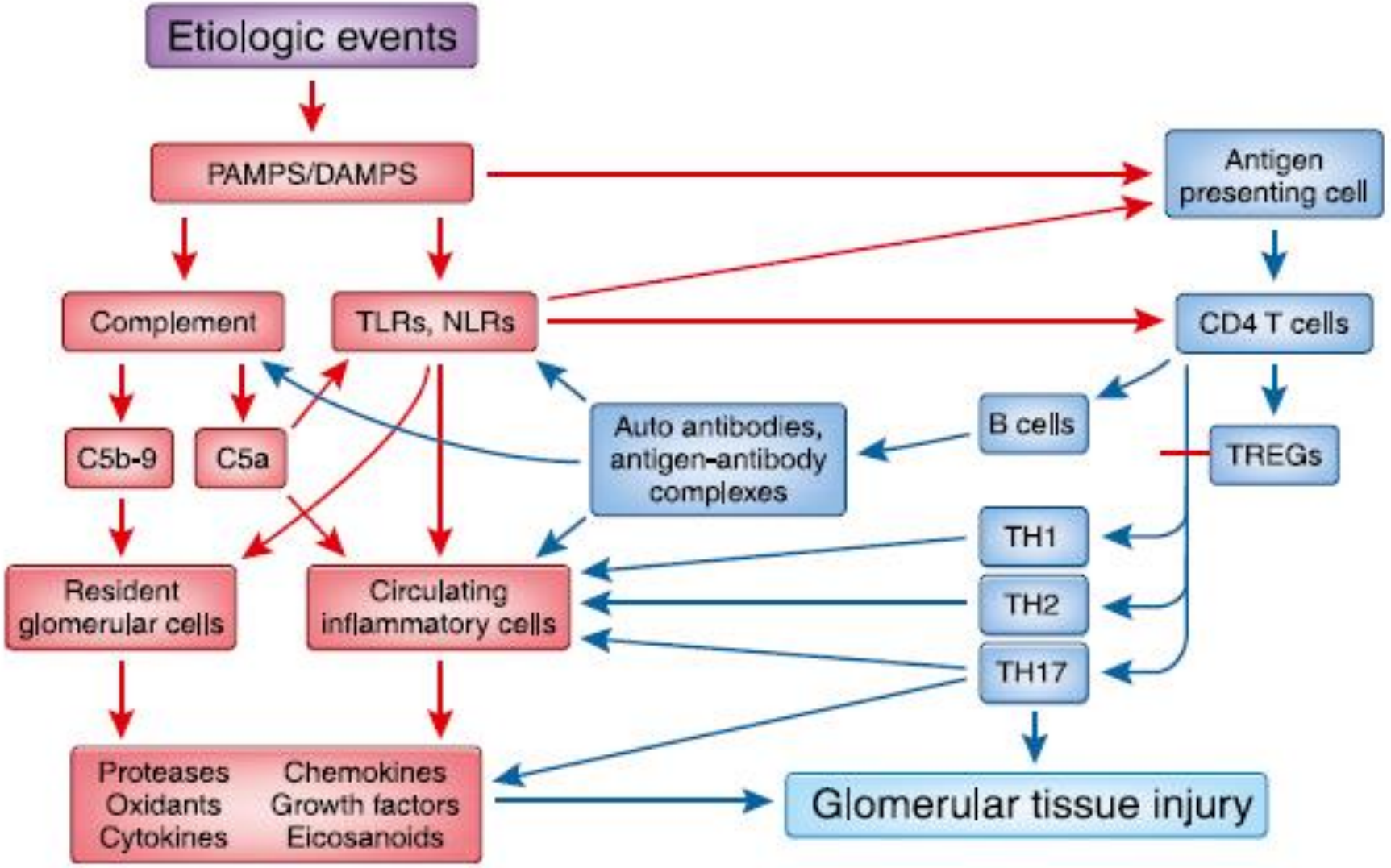
- Levels of plasma IgA1 are **elevated** in about half of the patients with IgAN as a result of an increased production of this isotype by the bone marrow and by a **low elimination rate** of the liver.
- The human IgA1 molecule is one of the few serum glycoproteins which possesses O-linked oligosaccharide (O-glycan) side chains in its hinge region.
- Altered O-glycosylation might favor **self-aggregation of IgA1** and **formation** of **circulating** IgA-containing ICs.
- There is also evidence for increased binding of abnormally O-glycosylated IgA1 to **extracellular matrix** (ECM) components.

- IgA receptor(FcαRI/CD89) is expressed on myeloid cells.
- Cross-linking of CD89 triggers various processes, including
- **phagocytosis**
- **superoxide generation**
- **antibody-dependent cellular cytotoxicity**
- **release of inflammatory mediators.**
- FcαRI/CD89 receptor has been reported to play a crucial role in the clearance of IgA complexes

Diagnostic of IgA nephropathy

- Clinical presentation
- Renal biopsy with immunofluorescence/ immunohistochemistry is the cornerstone of the diagnosis.
- Urine hematuria with or without proteinuria
- Serum polymeric IgA1: elevated in 30 - 50% of cases.
- Serum Gd-IgA1 and glycan specific autoantibody levels are prime candidates to become diagnostic biomarkers for IgA nephropathy.

Schematic overview of both the innate and adaptive immune mechanisms that mediate tissue injury in GN



Referense

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Thanks a lot